ECE 455 - Mathematical Models in Electrical Engineering

L2. Integrals of Complex Functions

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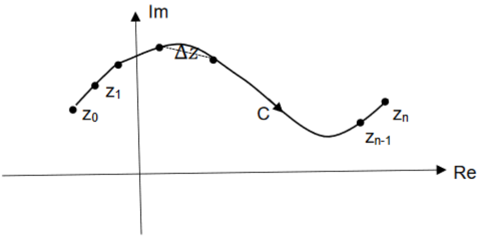
Outline

1. Line Integral
2. Cauchy-Goursat Integral Theorem
3. Applications of Cauchy-Goursat Integral Theorem
4. Cauchy's Integral Formulas

Line Integration

Line Integration



$$\int_C f(z)dz = \int_C udx - vdy + i \int_C vdx + udy \quad (2.3)$$

$$\oint_C f(z)dz = \oint_C udx - vdy + i \oint_C vdx + udy \quad (2.4)$$

Cartesian Method of Line Integration - Example

Example 1: Evaluate $\int_{0 \atop C}^{1+i2} \bar{z}^{\,2} dz$ where path C is shown in Figure 2.2.

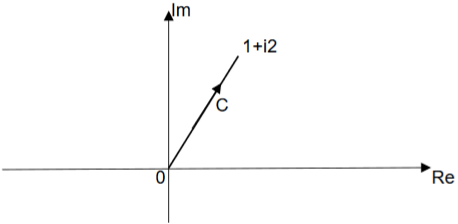


Figure 2.2 Path of Integration

Cartesian Method of Line Integration - Example (continued)

Example 1: Evaluate $\int_{0,\,C}^{1+i2} \bar{z}^{\,2} dz$ where path C is shown in Figure 2.2.

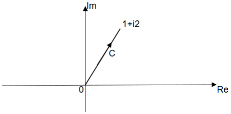


Figure 2.2 Path of Integration

Parametric Method of Line Integration

First we represent path C in parametric form z(t), unless it is already in that form.

$$C: \quad z(t) = x(t) + iy(t) \quad \text{or} \quad z(t) = r(t)\, e^{i\theta(t)} \qquad a \le t \le b \qquad (2.5)$$

Next we substitute z(t) in the integrand f(z(t)). Finally we write dz as

$$dz = \frac{dz(t)}{dt}\, dt \qquad (2.6)$$

where z(t) in equation (2.6) is the same as equation (2.5).

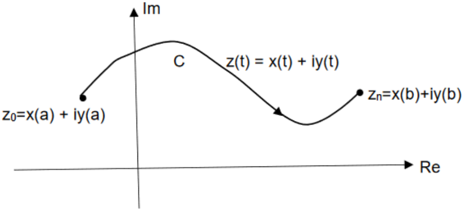


Figure 2.3 Parametric Path of Integration

The parametric form of equation (2.1) is

$$\underset{C}{\int_{z_0}^{z_n}} f(z)dz = \int_a^b f(z(t)) \frac{dz(t)}{dt}\, dt \qquad (2.7)$$

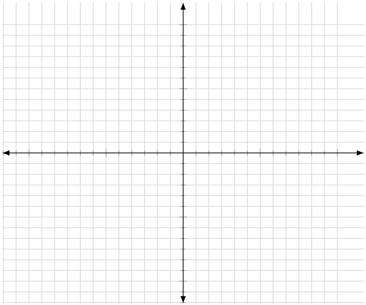
Curves Defined by Parametric Equations

This is a topic you should have covered in Calculus 3.
Most algebraic equations you have studied before this class were of the form $y = f(x)$ or as $x = g(y)$. Some equations cannot be written implicitly. Some curves are best written out in terms of a third variable, say t, called a parameter. For example, a point on a cartesian coordinate can be written as $(x = f(t), y = g(t))$.

Curves Defined by Parametric Equations

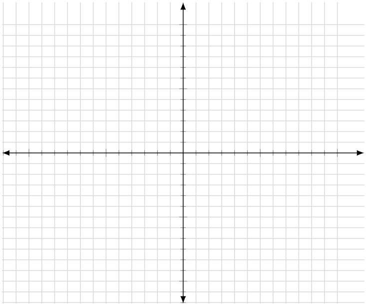
Example - Parametric Curves

Write the parametric representation for a line from $z = 0$ to $z_1 = x_1 + iy_1$.



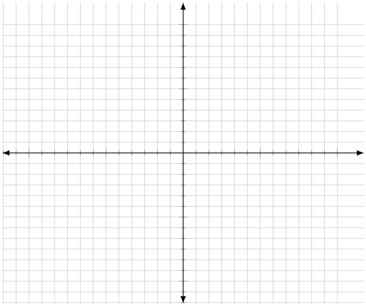
Example - Parametric Curves

Write the parametric representation for a line from $z_1 = x_1 + iy_1$ to $z_2 = x_2 + iy_2$.



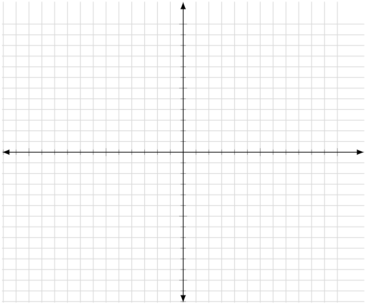
Example - Parametric Curves

Write the parametric representation for a circle centered at 0 with a radius ρ, in the counterclockwise direction.



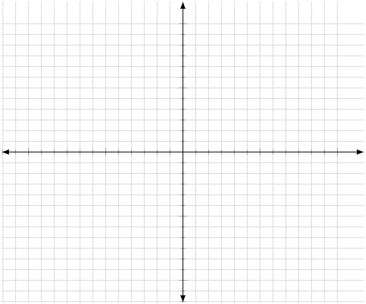
Example - Parametric Curves

Write the parametric representation for a semicircle centered at $z_0 = x_0 + iy_0$ with a radius ρ, in the counterclockwise direction.



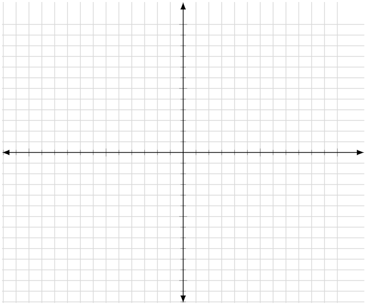
Example - Parametric Curves

Write the parametric representation for an ellipse centered at $z = 0$ whose cartesian equation is $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$



Example - Parametric Curves

Write the parametric representation for an ellipse centered at $z = h + ik$ whose cartesian equation is $\frac{(x-h)^2}{a^2} + \frac{(y-k)^2}{b^2} = 1$



Example

Evaluate $\int_{0\atop C}^{1+i2} \bar{z}^{\,2} dz$ where path C is shown in Figure 2.3.

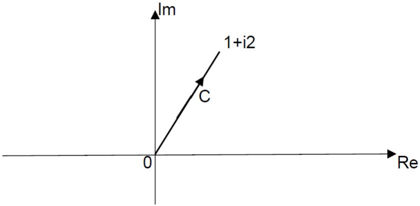


Figure 2.10 Path of Integration

Parametric Curves

If curve C is represented by $y = f(x)$, it suffices to represent the parametric form of this curve as,

C: $z(t) = t + if(t)$

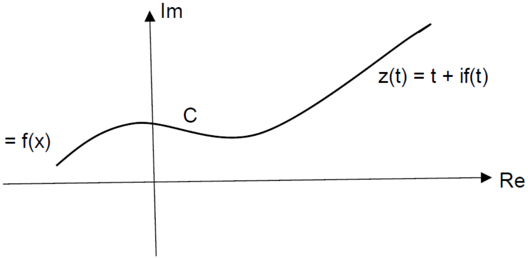


Figure 2.11 Parametric Path of Integration for $y = f(x)$

Example

$$\oint_C f(z)dz = \oint_C \frac{1}{(z-z_0)^n}dz$$

In the above equation, n is an integer and C is a closed circle of radius ρ with center at z_0 in counterclockwise direction.

ML Inequality

Let function f(z) be a continuous function for all values of z on curve C and let $|f(z)| \leq M$. Let the length C from initial to final point of integration be L. Then

$$\left|\int_C f(z)dz\right| \leq ML \tag{2.11}$$

Properties of Line Integrals

a) $\int_C kf(z)dz = k\int_C f(z)dz$ K is a complex constant (2.12)

b) $\int_C [f_1(z) \pm f_2(z)]dz = \int_C f_1(z)dz \pm \int_C f_2(z)dz$ (2.13)

c) $\int_{z_1}^{z_3} f(z)dz = \int_{z_1}^{z_2} f(z)dz + \int_{z_2}^{z_3} f(z)dz$ (2.14)

$C \qquad C_1 \qquad C_2$

Properties of Line Integrals

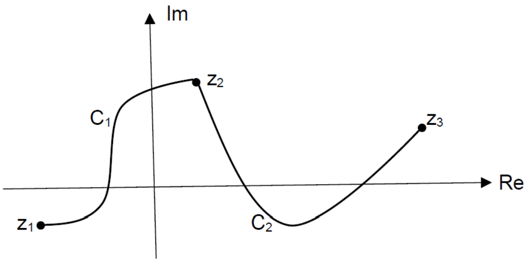


Figure 2.12 Curve C Consisting of partitions C_1 and C_2

$$\text{d)} \int_{z_1 \atop C_1}^{z_2} f(z)dz = -\int_{z_2 \atop C_2}^{z_1} f(z)dz \qquad (2.15)$$

Example

Evaluate $\int_{0}^{i} z \, dz$, where C consists of straight horizontal line from z = 0 to z = 1
C
and a quarter circle from z = 1 to z = i as shown.

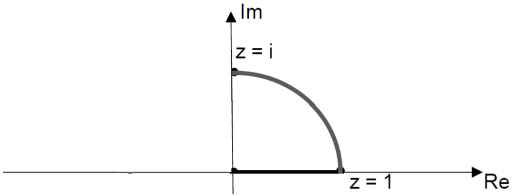


Figure 2.13

Example - Continued

Evaluate $\int_0^i z\, dz$, where C consists of straight horizontal line from z = 0 to z = 1
C
and a quarter circle from z = 1 to z = i as shown.

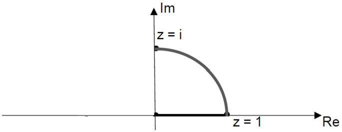


Figure 2.13

Example

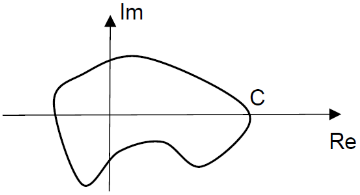
Evaluate $\int_0^i z\,dz$, where C consists of straight vertical line from z = 0 to z = i.
C

C: z(t) = i t $0 \leq t \leq 1$ dz/dt = i

$\int_0^i z\,dz$ = $\int_0^1 it(i)\,dt$ = -(1/2)
C

Cauchy-Goursat Integral Theorem

Simple and Not Simple Closed Paths



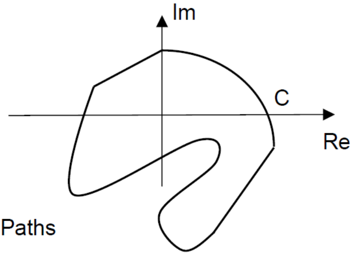
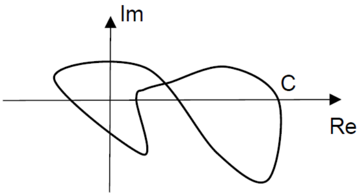


Figure 2.14 Simple Closed Paths



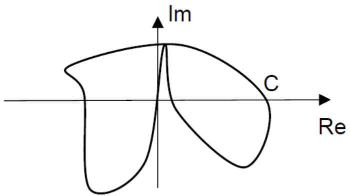


Figure 2.15 Not Simple Closed Paths

Simply and Multiply Connected Domain

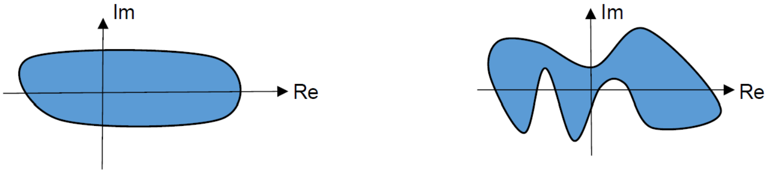


Figure 2.16 Simply Connected Domain

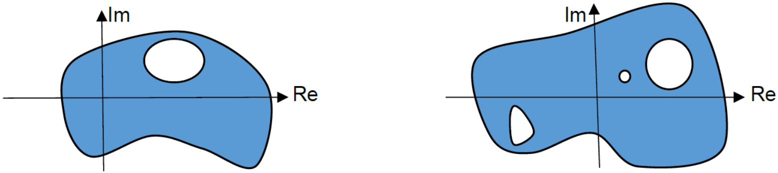


Figure 2.17 Not Simply Connected Domain

Cauchy-Goursat Integral Theorem

If the complex function f(z) is analytic everywhere in a simply connected domain D and C is any simple closed path C in D, then

$$\oint_C f(z)dz = 0 \qquad (2.16)$$

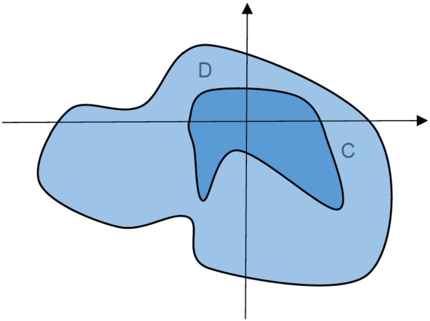


Figure 2.18 Simple Closed Curve in a Simply Connected Domain

Example

Example 7: Evaluate $\oint_C f(z)dz$ for the given f(z) and the closed path C in CCW direction

a) $f(z) = z^2 + 4z + e^z + 10$ C: $|z - i2| = 10$

Example

Example 7: Evaluate $\oint_C f(z)dz$ for the given f(z) and the closed path C in CCW

b) f(z) = $\dfrac{z+5}{(z-5)(z^2+9)}$

C:

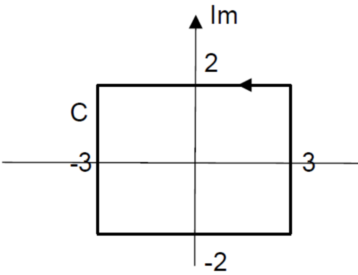


Figure 2.19 Path C – Examp

Example - Continued

<u>Example 7:</u> Evaluate $\oint_C f(z)dz$ for the given f(z) and the closed path C in CCW direction

b) $f(z) = \dfrac{z+5}{(z-5)(z^2+9)}$

C:

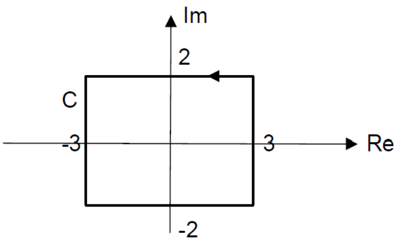


Figure 2.19 Path C – Example 7b

Applications of Cauchy-Goursat Integral Theorem

Partitioned Closed Path

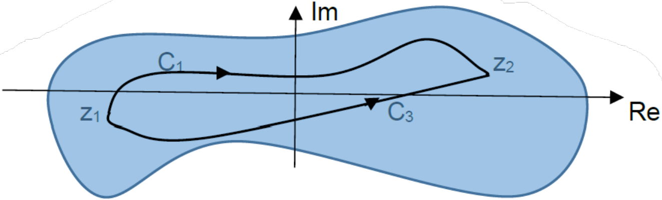


Figure 2.21 Closed Path C – Partitioned

Existence of Anti-derivative

If $f(z)$ is an analytic function everywhere in a simply connected domain D, then

$$\int_{z_1}^{z_2} f(z)\, dz = F(z_2) - F(z_1) \tag{2.20}$$

where $F(z)$ defined as antiderivative of $f(z)$ is an analytic function and $F'(z) = f(z)$ in D.

Example

Example 8: Evaluate $\int_{z_1}^{z_2} \mathrm{f(z)}dz$ for the given f(z) and path C.
C

a) $f(z) = z^2 + 2$ Path C consists of C_1 and C_2 as shown below

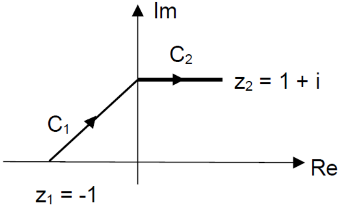


Figure 2.22

Example - Continued

<u>Example 8:</u> Evaluate $\int_{z_1 \atop C}^{z_2} f(z)dz$ for the given f(z) and path C.

a) $f(z) = z^2 + 2$ Path C consists of C_1 and C_2 as shown below

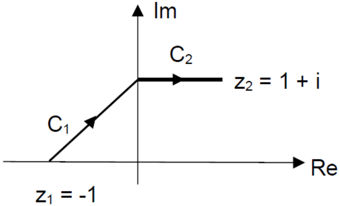


Figure 2.22

Example

Example 8: Evaluate $\int_{z_1}^{z_2} f(z)dz$ for the given f(z) and path C.
C

b) $f(z) = \sin 2z$ Path C as shown below

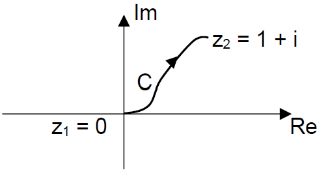


Figure 2.23

Example - Continued

<u>Example 8:</u> Evaluate $\int_{z_1}^{z_2} f(z)dz$ for the given f(z) and path C.
C

b) f(z) = sin 2z Path C as shown below

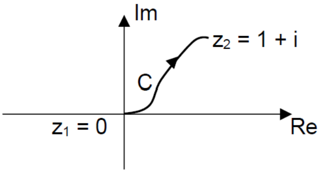


Figure 2.23

Multiply Connected Domain

$$\oint_{C} f(z)dz = 0$$

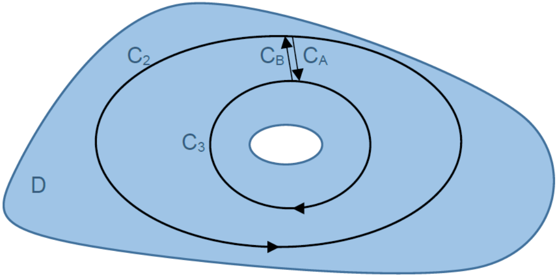


Figure 2.24 Doubly Connected Domain

Multiply Connected Domain

$$\oint_C f(z)dz = 0$$

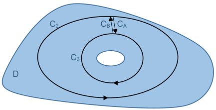


Figure 2.24 Doubly Connected Domain

Multiply Connected Domain

$$\oint f(z)dz = 0$$

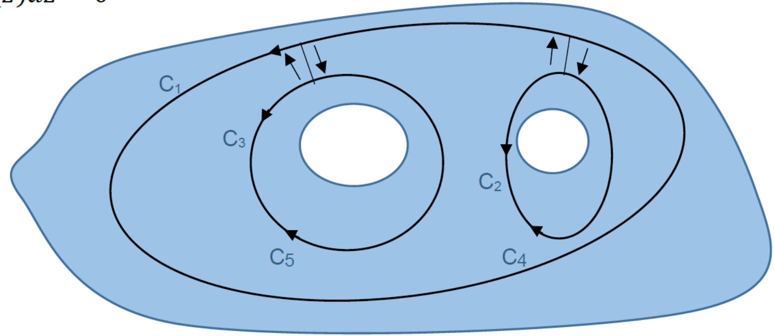


Figure 2.27 Triply Connected Domain

Multiply Connected Domain

$\oint f(z)dz = 0$

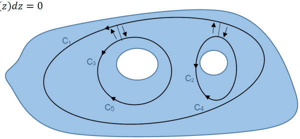


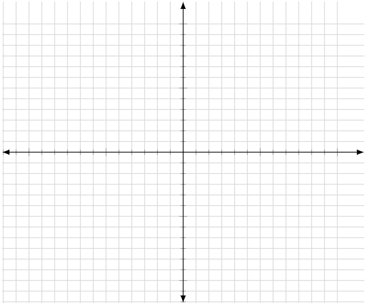
Figure 2.27 Triply Connected Domain

Example

Example 9: Evaluate $\oint_C f(z)dz$ for the given f(z) and path C.

$$f(z) = \frac{Z+10}{Z^3+4Z^2-5Z}$$

C: |z| = 4 in counterclockwise direction



Example - Continued

Example 9: Evaluate $\oint_C f(z)dz$ for the given f(z) and path C.

f(z) = $\dfrac{Z+10}{Z^3+4Z^2-5Z}$ C: |z| = 4 in counterclockwise direction

Cauchy's Integral Formulas

Cauchy's Integral Formula

If f(z) is an analytic function everywhere in a simply connected domain D and C is a simple closed path in D, then

$$\oint_C \frac{f(Z)}{Z-Z_0}\,dz = 2\pi i\, f(z_0) \qquad (2.28)$$

where z_0 is any point inside path C and integration is in counterclockwise direction.

Example

Example 10: Evaluate $\oint_C \frac{z^2+2}{(z^2+1)(z^2-1)} dz.$

C: $|z - i| = 1$ in counterclockwise direction

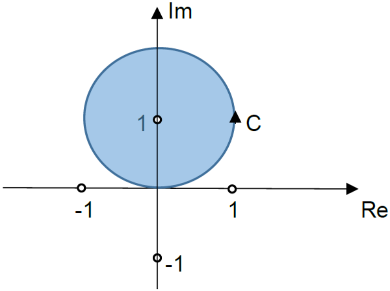


Figure 2.30

Cauchy's Integral Formula for Derivatives of Analytic Functions f(z)

If f(z) is an analytic function everywhere in a simply connected domain D and C is a simple closed path in D, then

$$f^{(n)}(z_0) = \frac{n!}{2\pi i} \oint_C \frac{f(Z)}{(Z-Z_0)^{n+1}}\, dz \qquad (2.33)$$

where z_0 is any point inside path C and integration is in counterclockwise direction.

Example

<u>Example 11:</u> Evaluate $\oint_C \frac{z^2+2z+2}{(z^2+4)(z-2)^2} dz$ for the given f(z) and path C.

C: As shown in Figure 2.31

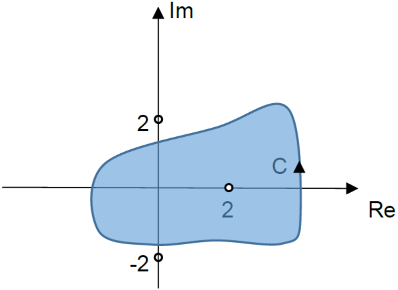


Figure 2.31

Liouville's Theorem

If f(z) is bounded and is an entire function for all values of z, then f(z) is constant.